HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2023)

Student No.: 15 Name: _____ Marks: / 42

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. If $y = 73.8$ (correct to 3 significant figures), find the range of values of $y$.

   A. $73.7 \leq y < 73.9$

   B. $73.7 < y \leq 73.9$

   C. $73.75 \leq y < 73.85$

   D. $73.75 < y \leq 73.85$

(DSE-2023 PII Q07)

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2. The scale of a map is $1 : 50000$. If the actual area of an airport is $10\,km^2$, then the area of this airport on the map is

A. $2\,cm^2$.

B. $4\,cm^2$.

C. $20\,cm^2$.

D. $40\,cm^2$.

(DSE-2023 PII Q12)

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3. Let $a_n$ be the $n$th term of a sequence. If $a_6 = 23$, $a_8 = 60$ and $a_{n+2} = a_{n+1} + a_n$ for any positive integer $n$, then $a_3 =$

A. 4.

B. 5.

C. 9.

D. 14.

(DSE-2023 PII Q14)

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4. The length of a side of a solid cube is $60\,\text{cm}$. The volume of a solid right circular cylinder is equal to the volume of the cube while the curved surface area of the circular cylinder is equal to the total surface area of the cube. Find the base radius of the circular cylinder.

A. $20\,\text{cm}$

B. $30\,\text{cm}$

C. $76\,\text{cm}$

D. $172\,\text{cm}$

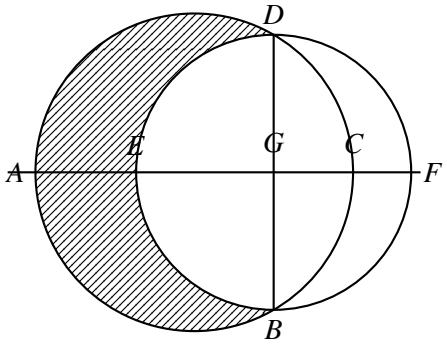
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5. In the figure, $AC$ is a diameter of the circle $ABCD$ while $BD$ and $EF$ are diameters of the circle $BEDF$. It is given that $C$ and $E$ lie on $AF$. Let $G$ be the point of intersection of $AF$ and $BD$. If $AG = 30\,\text{cm}$ and $CG = 10\,\text{cm}$, find the area of the shaded region correct to the nearest $\text{cm}^2$.




A. $209\,\text{cm}^2$

B. $367\,\text{cm}^2$

C. $383\,\text{cm}^2$

D. $540\,\text{cm}^2$

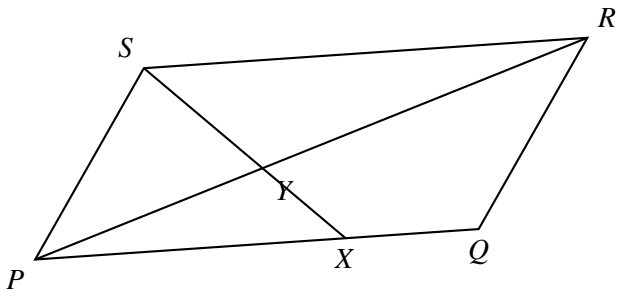
(DSE-2023 PII Q16)

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6. In the figure, $PQRS$ is a parallelogram. Let $X$ be a point lying on $PQ$. Denote the point of intersection of $PR$ and $SX$ by $Y$. If the area of $\triangle PXY$ and the area of the quadrilateral $QRYX$ are $32\,cm^2$ and $58\,cm^2$ respectively, then the area of $\triangle RSY$ is




A. $40\,cm^2$.

B. $50\,cm^2$.

C. $58\,cm^2$.

D. $72\,cm^2$.

(DSE-2023 PII Q17)

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7. It is given that $ABCD$ is a rhombus. Denote the point of intersection of $AC$ and $BD$ by $E$. Which of the following must be true?

I. $AE = BE$

II. $\frac{AE}{AC} = \frac{BE}{BD}$

III. $AE^2 + BE^2 = CD^2$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

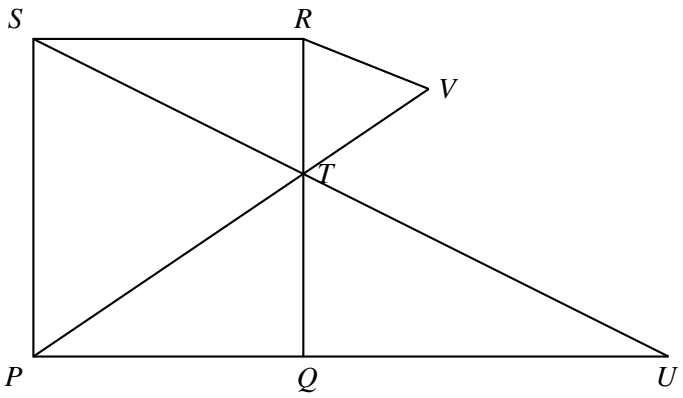
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8. In the figure, $PQRS$ is a rectangle. Let $T$ be a point lying on $QR$ such that $\angle PTS = 90°$. $PQ$ produced and $ST$ produced meet at the point $U$. $PT$ is produced to the point $V$ such that $RT = RV$. Which of the following must be true?




A. $RV \parallel ST$

B. $\angle PTQ = \angle RTS$

C. $\triangle PST \sim \triangle UTQ$

D. $\triangle PQT \cong \triangle TRS$

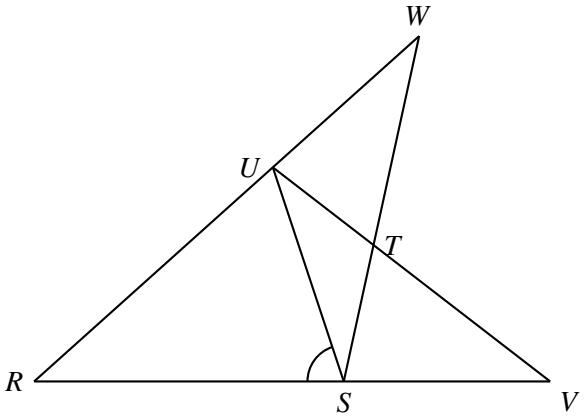
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9. The figure shows the cyclic quadrilateral $RSTU$, where $ST = TU$. $RS$ produced and $UT$ produced meet at the point $V$ while $RU$ produced and $ST$ produced meet at the point $W$. If $\angle RWS = 32^\circ$ and $\angle RVU = 48^\circ$, then $\angle RSU =$




A. $65^\circ$.

B. $73^\circ$.

C. $80^\circ$.

D. $82^\circ$.

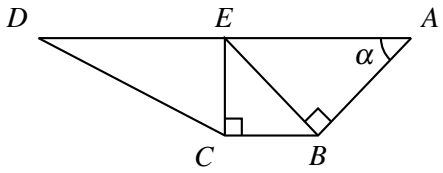
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10. In the figure, $ABCD$ is a trapezium with $AD//BC$. Let $E$ be the mid-point of $AD$. It is given that $\angle ABE = \angle BCE = 90°$. Find $\dfrac{CE}{DE}$.




A. $\dfrac{1}{2}$

B. $1$

C. $\tan\alpha$

D. $\sin\alpha\cos\alpha$

(DSE-2023 PII Q23)

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11. Find the constant $a$ such that the straight lines $2x+(a+3)y-5=0$ and $ax-4y+1=0$ are perpendicular to each other.

A. $-6$

B. $-5$

C. $-2$

D. 4

(DSE-2023 PII Q25)

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12. The equations of the straight lines $\ell$ and $L$ are $9x+12y-37=0$ and $12x+16y+85=0$ respectively. $\ell$ cuts the $x$-axis at the point $A$ while $L$ cuts the $y$-axis at the point $B$. Let $P$ be a moving point in the rectangular coordinate plane such that the perpendicular distance from $P$ to $\ell$ is equal to the perpendicular distance from $P$ to $L$. Denote the locus of $P$ by $\Gamma$. Which of the following are true?

I. $\Gamma$ is parallel to $L$.

II. $\Gamma$ is perpendicular to $AB$.

III. $\Gamma$ passes through the mid-point of $AB$.

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2023 PII Q26)

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13. A box contains five cards numbered 1 , 2 , 3 , 4 and 5 respectively while another box contains four cards numbered 6 , 7 , 8 and 9 respectively. If a number is randomly drawn from each box, find the probability that the product of the two numbers drawn is divisible by 4 .

A. $\frac{1}{5}$

B. $\frac{3}{10}$

C. $\frac{7}{20}$

D. $\frac{9}{20}$

(DSE-2023 PII Q28)

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14. The L.C.M. of $a^4b^2c$, $a^3b^4c$ and $a^2b^5c^2$ is

A. $a^2b^2c$.

B. $a^2b^2c^2$.

C. $a^4b^5c$.

D. $a^4b^5c^2$.

(DSE-2023 PII Q32)

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15. Let $f(x) = 3x^2 + 18mx + 22m^2$, where $m$ is a real constant. Which of the following statements about the graph of $y = -f(3x)$ must be true?

I. The $x$-coordinate of the vertex of the graph is $m$.

II. The $y$-coordinate of the vertex of the graph is $5m^2$.

III. The equation of the axis of symmetry of the graph is $x + m = 0$.

A. I only

B. II only

C. I and III only

D. II and III only

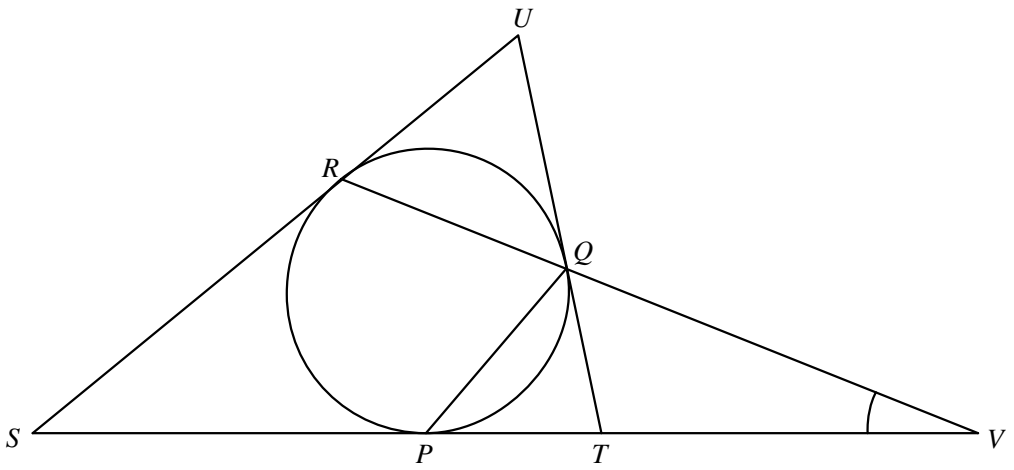
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16. In the figure, $P$, $Q$ and $R$ are points lying on a circle. $ST$, $TU$ and $SU$ are the tangents to the circle at $P$, $Q$ and $R$ respectively. $RQ$ produced and $ST$ produced meet at the point $V$. If $\angle PSR = 34^\circ$ and $\angle QPT = 46^\circ$, then $\angle PVQ =$




A. $17^\circ$.

B. $22^\circ$.

C. $27^\circ$.

D. $28^\circ$.

(DSE-2023 PII Q38)

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17. The base of the right pyramid $VABCD$ is the square $ABCD$. Let $\theta$ be the angle between $\triangle ABV$ and $\triangle BCV$. If $AB : AV = 5 : 4$, then $\cos\theta =$

A. $\dfrac{-25}{39}$.

B. $\dfrac{-17}{33}$.

C. $\dfrac{-9}{16}$.

D. $0$.

(DSE-2023 PII Q40)

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18. The equations of the straight lines $L_1$ and $L_2$ are $3x-4y+k=0$ and $4x+3y-k=0$ respectively, where $k$ is a positive constant. It is given that $L_1$ cuts the $x$-axis at the point $P$. Denote the point of intersection of $L_1$ and $L_2$ by $Q$. If $R$ is a point lying on $L_2$ such that the in-centre of $\triangle PQR$ lies on the $x$-axis, then the $x$-coordinate of $R$ is

A. $-7k$.

B. $-k$.

C. $k$.

D. $7k$.

(DSE-2023 PII Q41)

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19. There are 15 teachers in a group. If 5 teachers are selected from the group to form a committee consisting of 1 chairperson and 4 members, how many different committees can be formed?

A. 3003

B. 15015

C. 20475

D. 360360

(DSE-2023 PII Q42)

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20. When a boy throws a dart, the probability that he hits the target is $0.6$ . If this boy throws the dart 4 times, find the probability that he hits the target at least 2 times.

A. 0.5248

B. 0.7056

C. 0.8208

D. 0.8464

(DSE-2023 PII Q43)

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21. It is given that $n$ is an integer. Let $u$, $v$ and $w$ be the standard deviation, the median and the range of the group of numbers $\{1-9n, 3-9n, 4-9n, 5-9n, 7-9n\}$ respectively. Which of the following must be true?

I. $u = 2$

II. $v < 4$

III. $w > 6$

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2023 PII Q45)

I can finish _____ questions

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